

Topic

- Why model comparison is needed
 - Using Cross Validation properly
 - Comparing multiple models
 - Selecting best model
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Class Objective

Students will:

- Understand why one model is not enough
 - Learn how to **compare multiple models fairly**
 - Use **Cross Validation for reliable evaluation**
 - Select the **best performing model**
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1. CLASS START (Hook)

Start like this:

“We built multiple models — Logistic, Decision Tree, KNN...
But which one is actually best?”

 Students think → perfect entry

2. WHY MODEL COMPARISON?

Problem

- Different models → different results
 - Accuracy changes with data
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Example

Model	Accuracy
Logistic	82%
Decision Tree	90%
KNN	87%

 Which one to choose?

Teaching Line

“We don’t assume best model — we prove it.”

3. PROBLEM WITH NORMAL ACCURACY

Issue

- Train-test split depends on data
 - Results may change
-

 Already discussed earlier (CV)

Solution

 Use **Cross Validation**

4. USING CROSS VALIDATION PROPERLY

Idea

- Evaluate model multiple times
 - Take average score
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 **Teaching Line**

“Cross Validation gives stable and reliable performance.”

5. WHAT WE WILL DO TODAY

 Compare 3 models:

- Logistic Regression
- Decision Tree
- KNN

Using:

- `cross_val_score()`
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6. INTERVIEW CONCEPTS

Q1: Why compare models?

👉 To find best performing model.

Q2: Why use cross validation?

👉 More reliable than single split.

Q3: Can one model always be best?

👉 No, depends on data.

7. 🖥️ PRACTICAL (YOUR FORMAT)

🎯 Objective

Compare multiple models using Cross Validation

8. 📊 OUTPUT INTERPRETATION

Explain:

- Each model → multiple scores
 - Average score → final decision
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👉 Example:

Model	Avg Score
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Logistic	0.85
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Decision Tree	0.90
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KNN	0.88
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👉 Best = Decision Tree

“Best model is the one with highest average CV score.”

“A good data scientist never relies on one model — they compare and select the best one using proper evaluation.”

SESSION 17 – UNSUPERVISED LEARNING + K-MEANS CLUSTERING

🎯 Topic

- What is Unsupervised Learning
- Difference from Supervised Learning
- Clustering concept

- K-Means Algorithm
- Practical implementation

Class Objective

Students will:

- Understand **unsupervised learning concept**
- Learn how clustering works
- Understand **K-Means algorithm step-by-step**
- Perform **clustering practically**

1. CLASS START (Hook)

Start like this:

“Till now we always had output (Pass/Fail)...
But what if I give you data WITHOUT labels?”

👉 Pause → let them think

Then:

“Can we still find patterns? YES — that’s Unsupervised Learning.”

2. WHAT IS UNSUPERVISED LEARNING?

Definition

Unsupervised Learning is a type of ML where **no target/output column is given**.

Simple Meaning

“Model finds hidden patterns on its own.”

3. SUPERVISED vs UNSUPERVISED

Feature	Supervised	Unsupervised
Target column	Present	Not present
Example	Pass/Fail	Customer groups
Goal	Prediction	Pattern finding

Teaching Line

“Supervised learns with answers, Unsupervised finds answers.”

4. 📊 WHAT IS CLUSTERING?

Definition

Clustering groups similar data points together.

Example

- Students grouped by performance
 - Customers grouped by spending
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💡 Teaching Line

“Clustering = grouping similar things together.”

5. 📊 REAL-LIFE EXAMPLE (VERY IMPORTANT)

Mall Customers

Data:

- Age
- Income
- Spending Score

👉 Model groups customers into:

- High spenders
 - Low spenders
 - Medium
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👉 No labels given, still groups created!

6. 📖 INTRO TO K-MEANS

Definition

K-Means is a clustering algorithm that groups data into **K clusters**.

Simple Meaning

“K-Means divides data into K groups based on similarity.”

7. HOW K-MEANS WORKS

Step-by-Step

1. Choose K (number of clusters)
 2. Randomly select centroids
 3. Assign points to nearest centroid
 4. Update centroids
 5. Repeat until stable
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Teaching Line

“Clusters improve step-by-step until they stabilize.”

8. WHAT IS CENTROID?

Definition

Centroid = center of a cluster

Teaching Line

“Every cluster has a center point called centroid.”

9. IMPORTANT CONCEPT

 K-Means minimizes:

Distance between:

- Data points
 - Cluster center
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10. LIMITATIONS (IMPORTANT)

- ✗ Need to choose K manually
 - ✗ Sensitive to initial centroids
 - ✗ Not good for irregular shapes
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11. 🎯 INTERVIEW CONCEPTS

Q1: What is clustering?

👉 Grouping similar data.

Q2: What is K in K-Means?

👉 Number of clusters.

Q3: What is centroid?

👉 Center of cluster.

Q4: Does K-Means need labels?

👉 No.

12. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Group customers using K-Means

13. 📊 OUTPUT INTERPRETATION

Explain:

- Data divided into groups
 - Each color = cluster
 - Similar points grouped together
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👉 Example:

- Low income → low spend
- High income → high spend

“Unsupervised learning helps us discover hidden patterns in data without any labels — and K-Means is one of the simplest ways to do it.”